

SUPRIYA S

Florida USA • (904)-274-2321 • sup96478@gmail.com

PROJECT CASE STUDY

Real-Time + Batch Data Platform: Latency Optimization at Scale

Role: Data Engineer, Accenture 2022 - 2024 · Stack: AWS, Databricks, PySpark/Scala, Kafka, Snowflake, Redshift

CONTEXT

A client analytics platform ingested data from 12+ heterogeneous source systems (transactional DBs, weblogs, and event streams) into an Amazon S3 data lake. Pipelines were slow and brittle: high end-to-end latency delayed business reporting, and ad-hoc Spark jobs were expensive and hard to maintain. The goal was to deliver near real-time insight while improving reliability and cost.

APPROACH

Architecture: Sources → Kafka → Spark Streaming → S3 / Databricks (batch + streaming) → Snowflake / Redshift (serving)

- Re-architected ingestion into a unified batch + real-time design on AWS Glue, PySpark, and Databricks, replacing brittle point pipelines with a reusable, metadata-driven pattern.
- Integrated Kafka with Spark Streaming for low-latency event ingestion (~50K events/sec) into Cassandra and the lakehouse.
- Re-implemented performance-critical jobs in Scala for stronger type safety and JVM performance on web-scale datasets.
- Tuned partitioning, cluster configuration, and UDFs after profiling execution plans; added data-quality checks, monitoring, and lineage.
- Designed dimensional models (normalization/denormalization) and Spark SQL transformations across Snowflake and Redshift for efficient serving.

RESULTS

- **40% reduction** in end-to-end data-processing latency, enabling faster decision-making.
- **Critical job runtime cut from ~2 hours to ~45 minutes** through profiling and tuning.
- **~25% lower warehouse spend and query time** via improved modeling and transformations.
- **~30% fewer data defects** after embedding data-quality checks, monitoring, and lineage.
- **20+ reusable, documented pipelines** delivered from previously ambiguous requirements.